

INDIANAPOLIS INTL, IN, USA

WMO: 724380

Lat: 39.725N

Lon: 86.282W

Elev: 241

StdP: 98.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.6	-21.3	0.6	-16.2	-18.5	0.8	-12.9	12.6	0.4	11.7	-0.6	4.3	280	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	32.9	23.8	31.6	23.3	30.3	22.6	25.6	30.7	24.7	29.5	23.9	28.4	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.5	28.6	23.2	18.5	27.6	22.5	17.7	26.7	79.7	30.8	76.1	29.5	72.8	28.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.8	8.6	-20.2	34.6	4.4	1.8	-23.4	36.0	-26.0	37.0	-28.5	38.1	-31.7	39.4		
			-20.6	26.9	4.3	1.1	-23.7	27.6	-26.2	28.3	-28.6	28.8	-31.7	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-2.1	0.3	5.8	12.2	17.6	22.6	24.3	23.8	20.1	13.3	6.4	0.8
	DBStd	10.57	6.97	6.47	6.24	5.18	4.62	3.31	2.93	2.69	4.03	4.97	5.43	5.94
	HDD10.0	1303	378	277	162	36	3	0	0	0	23	134	291	
	HDD18.3	2902	633	506	391	193	71	7	1	28	170	358	544	
	CDD10.0	2088	4	6	32	101	239	377	443	428	303	125	27	4
	CDD18.3	644	0	0	2	8	49	133	186	171	81	13	1	0
	CDH23.3	5372	0	0	11	65	377	1079	1674	1412	654	98	1	0
	CDH26.7	1757	0	0	1	6	87	351	613	473	207	19	0	0
Wind	WSAvg	4.3	4.9	4.9	4.9	5.0	4.3	3.8	3.4	3.3	3.4	4.1	4.5	4.6
Precipitation	PrecAvg	1049	67	61	91	102	110	106	113	87	78	73	84	79
	PrecMax	1337	239	136	229	218	235	261	334	212	263	199	216	196
	PrecMin	779	9	9	16	25	27	2	12	9	6	1	18	11
	PrecStd	157	43	32	45	46	58	56	70	51	54	41	43	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.8	19.5	24.9	27.7	30.9	33.1	35.9	34.2	33.3	29.0	22.3	17.9
		MCWB	14.4	13.7	16.8	18.6	21.4	23.1	25.2	23.9	21.9	20.7	15.7	14.8
	2%	DB	13.9	15.7	21.6	25.3	28.9	31.6	33.0	32.3	31.2	26.1	19.4	14.6
		MCWB	12.1	12.2	14.9	16.9	20.7	23.0	24.6	23.8	21.8	18.8	14.7	12.3
	5%	DB	11.1	12.7	18.5	23.1	27.4	30.2	31.5	30.9	29.1	23.7	17.1	12.0
		MCWB	9.4	9.6	13.1	15.8	19.8	22.5	23.8	23.1	20.9	17.7	13.5	10.0
	10%	DB	7.7	9.8	15.6	20.9	25.4	28.8	30.0	29.3	27.1	21.5	14.9	9.1
		MCWB	5.8	6.8	11.3	14.8	19.2	21.7	23.0	22.1	20.0	16.6	11.6	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.6	17.8	20.5	23.5	25.7	26.9	26.3	24.3	22.4	17.6	15.4
		MCDB	16.4	18.5	23.3	25.6	28.3	31.0	32.8	31.5	29.6	26.8	20.7	17.6
	2%	WB	12.5	12.8	15.8	18.4	22.1	24.6	25.8	25.2	23.2	20.6	15.7	12.7
		MCDB	13.8	14.6	20.2	22.7	27.3	29.7	31.1	30.4	28.4	24.4	18.3	14.1
	5%	WB	9.9	10.0	14.1	17.0	21.1	23.7	25.0	24.3	22.3	18.9	14.1	10.2
		MCDB	11.0	12.7	17.5	21.2	25.7	28.5	29.8	28.8	26.9	22.5	16.6	11.7
	10%	WB	5.7	6.7	11.8	15.6	20.1	22.8	24.1	23.5	21.4	17.1	11.9	7.2
		MCDB	7.5	9.5	15.2	19.9	24.2	27.2	28.4	27.5	25.3	20.5	14.7	8.9
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	9.9	10.9	10.2	9.7	9.8	9.8	10.7	10.3	8.9	7.4
		MCDBR	10.3	12.5	13.1	13.0	11.8	11.0	10.8	11.0	12.5	12.6	11.4	9.9
		MCWBR	8.9	9.6	8.1	7.5	5.9	4.9	4.3	4.4	4.9	6.6	8.1	8.4
	5% WB	MCDBR	9.5	11.5	11.7	10.9	10.2	9.8	9.8	9.5	10.3	10.4	10.2	9.3
MCWBR		8.9	9.9	7.8	7.5	5.7	5.1	4.6	4.5	4.8	6.1	8.3	8.6	
Clear-Sky Solar Irradiance	taub	0.303	0.323	0.348	0.404	0.438	0.442	0.457	0.453	0.418	0.359	0.321	0.301	
	taud	2.437	2.357	2.370	2.223	2.177	2.206	2.166	2.164	2.243	2.410	2.466	2.459	
	Ebn at Noon	865	893	906	870	845	839	822	815	820	841	837	840	
	Edh at Noon	81	100	111	137	147	143	147	144	125	95	78	74	
All-Sky Solar Radiation	RadAvg	1.76	2.58	3.64	4.75	5.41	6.17	6.01	5.48	4.52	3.17	2.12	1.48	
	RadStd	0.19	0.29	0.32	0.39	0.49	0.37	0.35	0.32	0.37	0.30	0.23	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	-0.47	N/A	N/A	+101	N/A
Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	N/A

Nomenclature: See separate page